

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓	✓		✓
CO 4		✓		✓
CO 5		✓		✓
CO 6			✓	✓

Programme	BSc Statistics				
Course Code	ACT2MN101 (P)				
Course Title	Actuarial mathematics II				
Type of Course	Minor				
Semester	II				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Basic knowledge of present value, probability, mean and variance, Basic computer skills				
Course Summary	The aim of this course is to expose the students about Life and Health Contingencies.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	To demonstrate the life table functions and select mortality.	An	C	Instructor-created exams / Quiz
CO2	To explain the life assurance contract and benefits.	U	C	Instructor-created exams / Home

				Assignments
CO3	Develop formulae for the means and variances of the payments under various assurance and annuity contracts	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	Define various annuity contracts.	U	C	Instructor-created exams / Home Assignments
CO5	Summarize various Joint life and last survivor functions	R	F	One Minute Reflection Writing assignments
CO6	Solving life and health contingencies with sample data with spread sheet	Ap	P	Viva-Voce/Practical Assignment
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hours (45 +30)	Marks (70)
I	The life table		12	20
	1	Life table, Constructing a life table	3	
	2	The force of mortality, Lifetime random variables, The pattern of human mortality	2	
	3	Life table functions at non-integer ages, uniform distribution of deaths (UDD), constant force of mortality (CFM)	4	
	4	Evaluating probabilities without use of the life table	1	
	5	Select mortality, Mortality rates that depend on both age and duration	2	
	Sections from References: Unit 1: 15.0&15.2 (15.2.1 to 15.2.2) [Ref 1] Unit 2: 15.2.3,15.2.6,15.2.7 [Ref 1] Unit 3: 15.3 (15.3.1 to 15.3.3) [Ref 1] Unit 4: 15.4 [Ref 1] Unit 5: 15.5,15.5.1, 15.5.2 [Ref 1]			
II	Life assurance contracts		13	20

	6	Whole life assurance contracts, Term assurance contracts	4	
	7	Pure endowment contracts, Endowment assurance contracts, Deferred assurance benefits	4	
	8	Benefits payable immediately on death (Whole life assurance, Term assurance, Endowment assurance)	3	
	9	Evaluating means and variances using select mortality.	2	
	Sections from References: Unit 6: 16.1 (16.1.1 to 16.1.3), 16.2 (16.2.1 to 16.2.3) [Ref 1] Unit 7: 16.3(16.3.1 to 16.3.3), 16.4 (16.4.1 to 16.4.3), 16.5(16.5.1, 16.5.2) [Ref 1] Unit 8: 16.6(16.6.1 to 16.6.4) - [Ref 1] Unit 9: 16.7- [Ref 1]			
III	Life annuity contracts		11	15
	10	Life annuity contracts, Whole life annuities payable annually in arrears	2	
	11	Whole life annuities payable annually in advance	1	
	12	Temporary annuities payable annually in arrears	1	
	13	Temporary annuities payable annually in advance	2	
	14	Deferred annuities, Deferred annuities-due	3	
	15	Continuous annuities , Other annuities	2	
	Sections from References: Unit 10: 17.1,17.2(17.2.1 to 17.2.3) [Ref 1] Unit 11: 17.3 (17.3.1 to 17.3.2) [Ref 1] Unit 12: 17.4 (17.4.1 to 17.4.2) [Ref 1] Unit 13: 17.5(17.5.1 to 17.5.2) [Ref 1] Unit 14: 17.6(17.6.1 to 17.6.2), 17.7 [Ref 1] Unit 15: 17.10,17.10.1, 17.10.2 [Ref 1]			
IV	Joint life and last survivor functions		9	15
	16	Random variables to describe joint life functions	1	
	17	Joint lifetime random variables and joint life table functions	2	
	18	Last survivor lifetime random variables	1	
	19	Simple probabilities involving two lives	1	
	20	Evaluating last survivor functions	1	
	21	Present values involving two lives	2	
	22	Present values of joint life and last survivor annuities	1	

	Sections from References: Unit 16: 22.1(22.1.1 to 22.1.2) [Ref 1] Unit 17: 22.1.3 [Ref 1] Unit 18: 22.1.4[Ref 1] Unit 19: 22.2,22.2.1 [Ref 1] Unit 20: 22.2.2 [Ref 1] Unit 21: 22.3 & 22.3.1[Ref 1] Unit 22: 22.3.2 [Ref 1]		
V	PRACTICUM	30	
	Do practice problems in spreadsheet from any 5 units of the given list and one additional problem decided by the teacher-in-charge, related to the content of the course. Other units listed here may be used as demonstrations of the concepts taught in the course.		
	1 Life table using ultimate mortality		
	2 Select mortality		
	3 UDD and CFM assumption		
	4 Mean and variance of whole life assurance		
	5 Term assurance and endowment assurance		
	6 Life annuity contracts		
	7 Joint life functions		
	8 Last survivor functions		
Books and References: 1. Institute of Actuaries Act Ed. Study materials CM1 2. Dickson, Mary R. Hardy and Howard R.Waters (2019), Actuarial Mathematics for Life Contingent Risks , Cambridge University Press, India 3. McCutcheon, J.J., Scott William (1986): An introduction to Mathematics of Finance 4. Butcher,M.V., Nesbit, Cecil. (1971)Mathematics of compound interest, Ulrich's Books 5. David Promislow, S. (2014), Fundamentals of Actuarial Mathematics, John wiley& sons. 6. Newton LBowers, et al (1997):Actuarial Mathematics, The Societies of Actuaries, 2nd Ed 7. Shailaja R. Deshmukh- Actuarial Statistics-an introduction using R, 3rd Ed.			

Mapping of COs with PSOs and POs:

	PSO 1	PSO 2	PSO 3	PSO4	PSO 5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	2	3	1	2	2	-	2	-	2	-	-	-
CO 2	3	3	1	2	1	-	3	1	2	3	-	-
CO 3	1	2	-	2	2	-	2	-	2	-	-	-
CO 4	3	2	1	1	-	-	3	-	2	2	-	-
CO 5	3	2	-	-	-	-	3	-	2	-	-	-

CO 6	1	1	2	-	3	3	2	2	1	-	2	3
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Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓	✓		✓
CO 4		✓		✓
CO 5		✓		✓
CO 6			✓	✓

Programme	BSc Statistics
Course Code	ACT3MN201 (P)
Course Title	Risk Modelling and Survival Analysis

Type of Course	Minor				
Semester	III				
Academic Level	200 - 299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Basic knowledge of life table, probability, Basic computer skills				
Course Summary	The aim of this course is to provide a grounding in mathematical and statistical modelling techniques that are of particular relevance to actuarial work, including survival models and their application.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Describe and apply techniques of survival analysis	Ap	P	Instructor-created exams / Quiz
CO2	Describe Apply compound distributions in risk modelling.	Ap	P	Instructor-created exams / Home Assignments
CO3	Understand the concept of reinsurance	U	C	Seminar Presentation / Group Tutorial Work
CO4	Memorize the basic concepts of ruin theory.	R	C	Instructor-created exams / Home Assignments
CO5	Understand the basics of Machine Learning	U	F	One Minute Reflection Writing assignments
CO6	Solving survival models with sample data with spread sheet	Ap	P	Viva-Voce/Practical Assignment
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hours (45 +30)	Marks (70)
I	Survival models-I		13	20
	1	Survival models, Future lifetime, Probabilities of death and survival	3	
	2	The force of mortality, Survival probabilities	1	
	3	The probability density function of Tx, Life table functions, Initial and central rates of mortality	3	
	4	Expected future lifetime, Complete expectation of life, Curtate expectation of life, Some important formulae	3	
	5	Simple parametric survival models, The Gompertz and Makeham laws of mortality, Survival probabilities	3	
	Sections from References: Unit 1: 6.1 (6.1.1 to 6.1.2) [Ref 1] Unit 2: 6.1.3&6.1.4 [Ref 1] Unit 3: 6.1.5,6.1.6,6.1.7 [Ref 1] Unit 4: 6.2,6.2.1,6.2.2, 6.3 [Ref 1] Unit 5: 6.4, 6.5, 6.5.2 [Ref 1]			
II	Risk Models- I		12	20
	6	General features of a product, Models for short-term insurance contracts	2	
	7	The collective risk model, The basic model, Notation and assumptions	1	
	8	Distribution functions and convolutions	2	
	9	Moments of compound distributions, The compound Poisson distribution,	3	
	10	The compound binomial distribution	2	
	11	The compound negative binomial distribution	2	
	Sections from References: Unit 6: 19.1(19.1.1 to 19.1.2), 19.2 [Ref 1] Unit 7: 19.3, 19.3.1 [Ref 1] Unit 8: 19.3.2 [Ref 1] Unit 9: 19.3.3,19.3.4 [Ref 1] Unit 10: 19.3.5 [Ref 1] Unit 11: 19.3.6 [Ref 1]			
III	Reinsurance & Risk Models- II		11	18
	12	Reinsurance and its types	1	
	13	Reinsurance arrangements, Excess of loss reinsurance(concept only), proportional reinsurance (concept only)	2	

	14	The individual risk model	2	
	15	Ruin theory, Basic concepts	2	
	16	The surplus process	2	
	17	The probability of ruin in continuous & discrete time	2	
	Sections from References: Unit 12: 18.0 [Ref 1] Unit 13: 18.1,18.1.1,18.1.3 [Ref 1] Unit 14: 20.2 [Ref 1] Unit 15: 8.1 [Ref 2] Unit 16: 8.1.2 [Ref 2] Unit 17: 8.1.3 & 8.1.4 [Ref 2]			
IV	Machine learning		9	12
	18	What is machine learning?	1	
	19	An overview of machine learning	2	
	20	Concepts in machine learning- The loss function, Model evaluation, Generalisation error and model validation, Train- validation- test	2	
	21	Branches of machine learning	2	
	22	Stages of Analysing Machine learning	2	
	Sections from References: Unit 18: 21.1 [Ref 1] Unit 19: 21.2 [Ref 1] Unit 20: 21.3(21.3.1 to 21.3.4)[Ref 1] Unit 21: 21.4 [Ref 1] Unit 22: 21.5 (21.5.1 to 21.5.9)[Ref 1]			
V	PRACTICUM		30	
	Do practice problems in spreadsheet from any 5 units of the given list and one additional problem decided by the teacher-in-charge, related to the content of the course. Other units listed here may be used as demonstrations of the concepts taught in the course.			
	1	Survival probabilities		
	2	Expectation of life		
	3	Laws of mortality		
	4	Life table functions		
	5	Compound Poisson distribution		
	6	Compound binomial distribution		
	7	Distribution function		
	8	Surplus process		

Books and References:

1. Institute of Actuaries Act Ed. Study materials CS2.
2. Institute of Actuaries Act Ed. Study materials CT6.

3. Denuit,M., Marechal, X., Pitrebois, S., Walhin, J.F. (2007). Actuarial Modelling of claim counts: Risk classification, credibility and bonus-malus systems. John Wiley & Sons
4. Bowers, Newton Let al (1997): Actuarial mathematics, society of Actuaries, 2nd Ed
5. Benjamin,B; Pollard, J.H. (1993).The analysis of mortality and other actuarial Statistics: (3rd Ed). Institute and faculty of Actuaries
6. Shailaja R. Deshmukh- Actuarial Statistics-an introduction using R, 3rd Ed.
7. Daykin C.D, Pentikainen T. , Pesonen M.: Practical Risk theory for Actuaries (1194). Chapman& Hall.

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CO 3	1	2	-	3	-	-	2	-	2	-	-	-
CO 4	3	2	-	2	1	-	3	-	2	2	-	-
CO 5	3	2	-	-	-	-	3	-	2	-	-	-
CO 6	1	1	2	-	3	3	2	2	1	-	2	2

Correlation Levels:

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